```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
%matplotlib inline
sns.set_style('whitegrid')
```

```
# Task 1

days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
cups = [120, 135, 128, 140, 180, 260, 240]

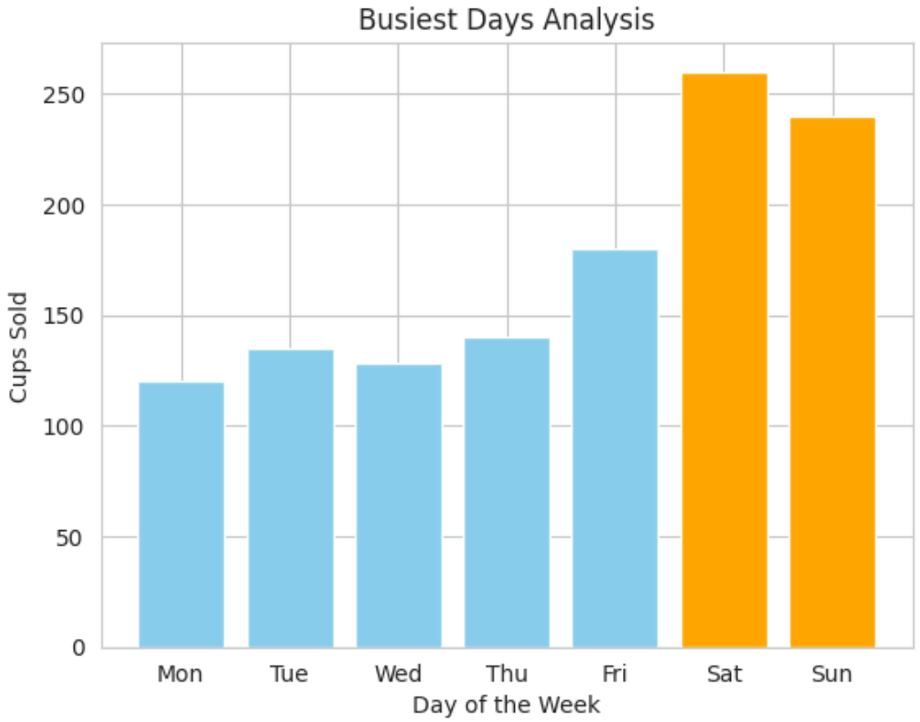
top = sorted(cups)[-2]
colors = ['orange' if x >= top else 'skyblue' for x in cups]


fig, ax = plt.subplots()

ax.bar(days, cups, color=colors)
ax.set_title('Busiest Days Analysis')
ax.set_xlabel('Day of the Week')
ax.set_ylabel('Cups Sold')

plt.show()

# The busiest days are Sat and Sun, so the owner should schedule more staff on weekends.
```




```
# Task 2

days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
steps = [6200, 8100, 7400, 9500, 11000, 3200, 5400]

daily_goal = 8000
missed_days = [day for day, x in zip(days, steps) if x < daily_goal]

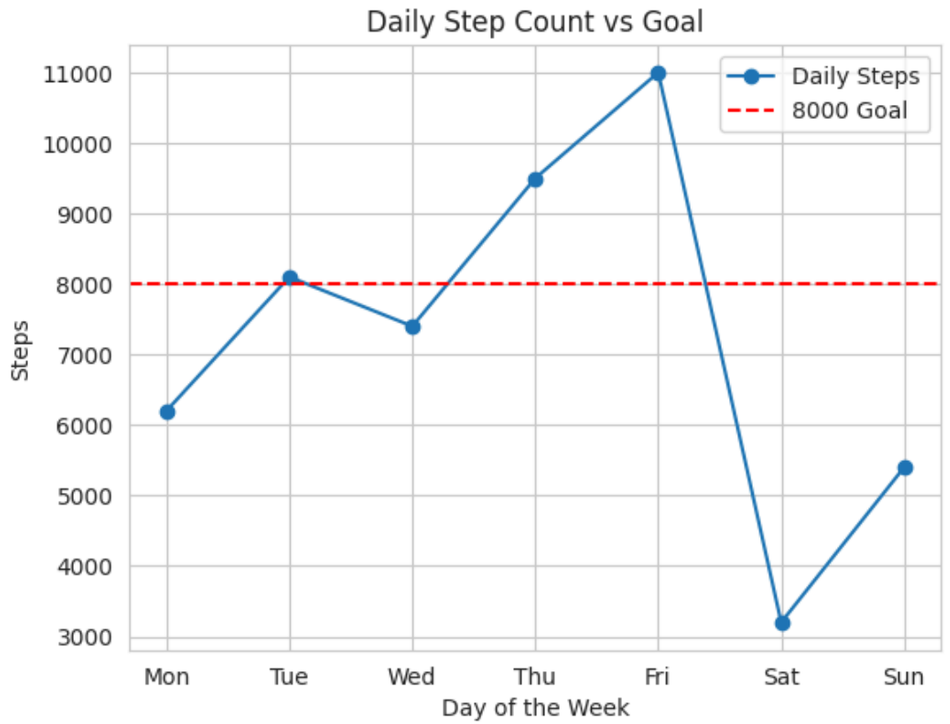
fig, ax = plt.subplots()

ax.plot(days, steps, marker='o', label='Daily Steps')
ax.axhline(y=daily_goal, color='red', linestyle='--', label='8000 Goal')

ax.set_title('Daily Step Count vs Goal')
ax.set_xlabel('Day of the Week')
ax.set_ylabel('Steps')
ax.legend()
```

```
plt.show()

# 🏆 So what? On which days did you MISS the goal?
# I missed the goal on Mon, Wed, Sat, and Sun.
```




```
# Task 3

genres = ['Drama', 'Comedy', 'Action', 'Docs', 'Horror', 'Sci-Fi']
hours = [42, 39, 36, 30, 28, 25]

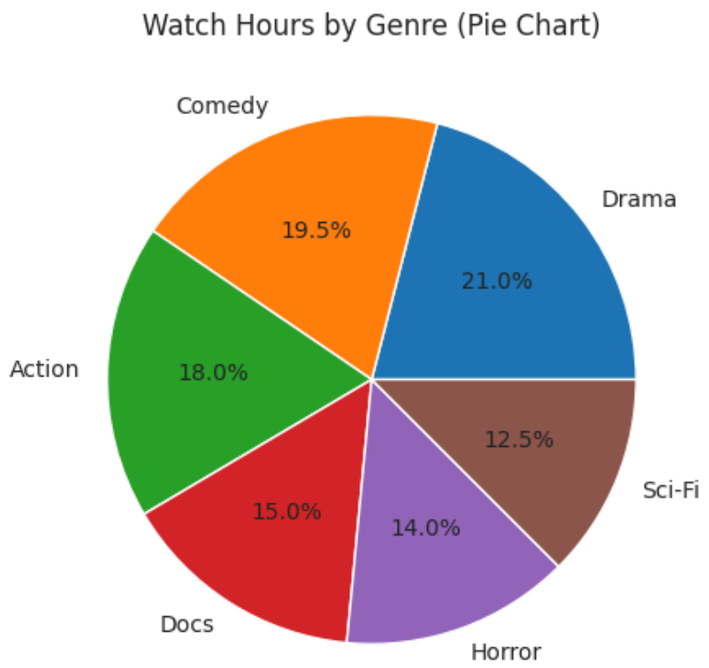

sorted_data = sorted(zip(hours, genres))
sorted_hours, sorted_genres = zip(*sorted_data)
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

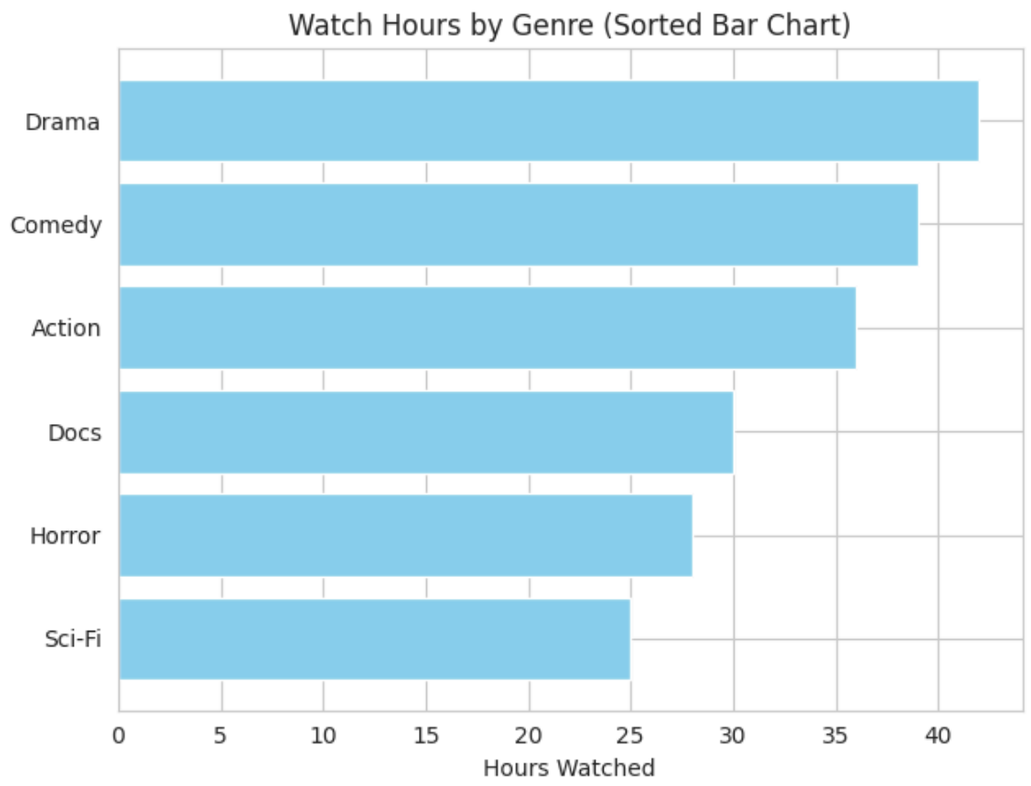
ax1.pie(hours, labels=genres, autopct='%1.1f%%')
ax1.set_title('Watch Hours by Genre (Pie Chart)')
```

```
ax2.barh(sorted_genres, sorted_hours, color='skyblue')
ax2.set_title('Watch Hours by Genre (Sorted Bar Chart)')
ax2.set_xlabel('Hours Watched')

plt.tight_layout()
plt.show()

# 🏆 So what? Which chart makes the ranking obvious, and why?
# The sorted horizontal bar chart makes the ranking obvious because bar lengths are easier to compare, clearly showing Drama at the top.
```






```
# Task 4

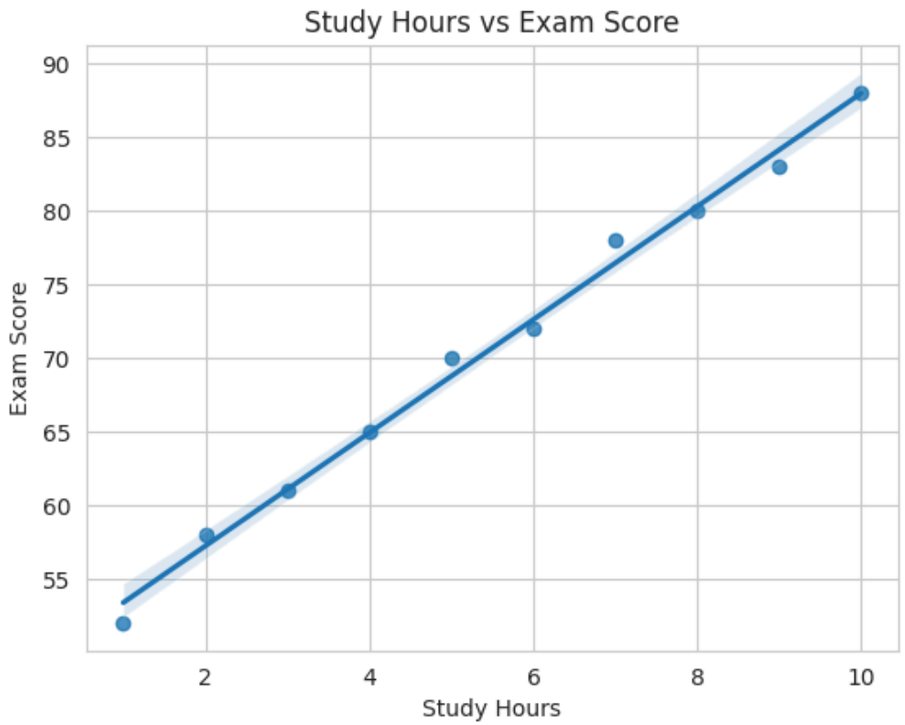
study_hours = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
exam_score = [52, 58, 61, 65, 70, 72, 78, 80, 83, 88]
```

```
sns.regplot(x=study_hours, y=exam_score)

plt.title('Study Hours vs Exam Score')
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')

plt.show()

# 🏆 So what? Is the relationship positive, negative, or none?
# The relationship is positive, as study hours increase, exam scores also increase.
```



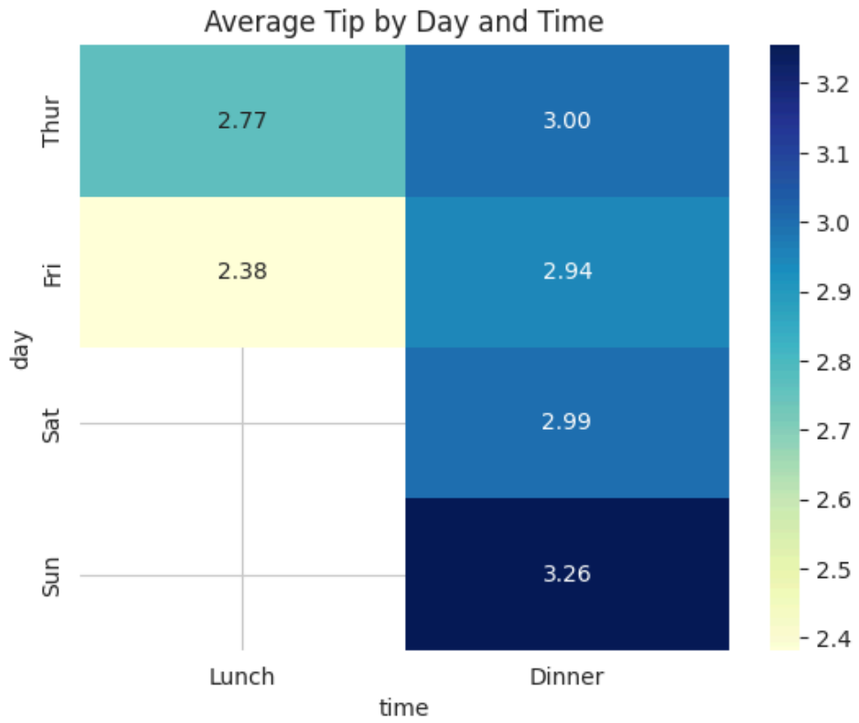

```
# Task 5

tips = sns.load_dataset('tips')

pivot_tips = tips.pivot_table(index='day', columns='time', values='tip', aggfunc='mean', observed=False)
sns.heatmap(pivot_tips, annot=True, cmap='YlGnBu', fmt='.2f')
```

```
plt.title('Average Tip by Day and Time')
plt.show()

# 🏆 So what? Which day + time has the highest average tip?
# Sunday Dinner has the highest average tip ($3.12).
```




```
# Task 6

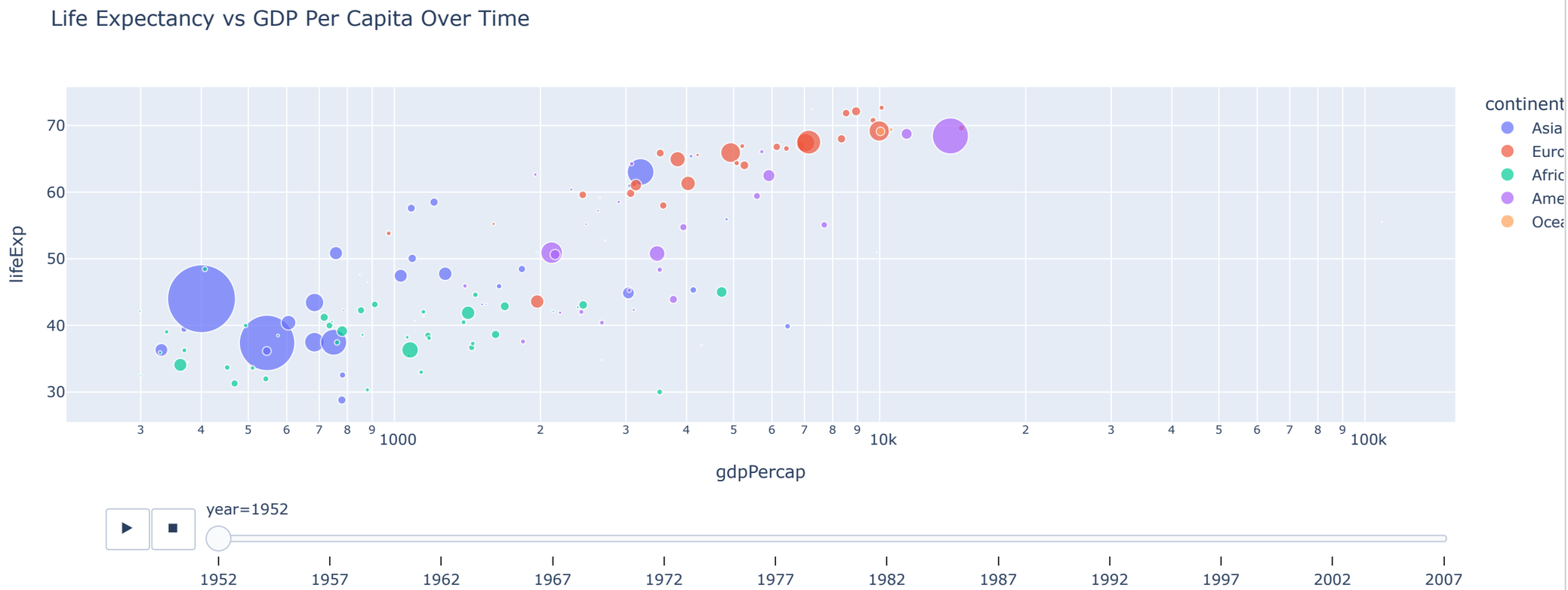
gapminder = px.data.gapminder()

fig = px.scatter(
    gapminder,
    x='gdpPercap',
    y='lifeExp',
    size='pop',
    color='continent',
    hover_name='country',
    log_x=True,
```

```
    size_max=60,
    animation_frame='year',
    title='Life Expectancy vs GDP Per Capita Over Time'
)

fig.show()

# 🏆 So what? What overall trend do you see as the years play forward?
# As the years play forward, both GDP per capita and life expectancy increase across most countries.
```




```
# Task 7

df = pd.DataFrame({
    'Day':     ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'],
```

```
    'Orders':  [  45,    62,    58,    70,    95,   110,    80 ],
    'Revenue': [ 900,  1240,  1160,  1400,  1900,  2200,  1600 ],
})

cat_names = ['Electronics', 'Clothing', 'Home', 'Books']
cat_revenue = [4200, 3100, 2000, 1100]

fig, axes = plt.subplots(2, 2, figsize=(13, 9))

# (0,0) Chart type chosen: Line Chart | Why: Shows continuous trend over time (days of the week)
axes[0, 0].plot(df['Day'], df['Revenue'], marker='o', color='teal')
axes[0, 0].set_title('Daily Revenue Trend')
axes[0, 0].set_xlabel('Day')
axes[0, 0].set_ylabel('Revenue ($)')

# (0,1) Chart type chosen: Bar Chart | Why: Best for comparing discrete categories (orders per day)
axes[0, 1].bar(df['Day'], df['Orders'], color='royalblue')
axes[0, 1].set_title('Orders Comparison Across Days')
axes[0, 1].set_xlabel('Day')
axes[0, 1].set_ylabel('Number of Orders')

# (1,0) Chart type chosen: Scatter Plot | Why: Best for showing relationship between two numeric variables
axes[1, 0].scatter(df['Orders'], df['Revenue'], color='crimson')
axes[1, 0].set_title('Orders vs Revenue Relationship')
axes[1, 0].set_xlabel('Orders')
axes[1, 0].set_ylabel('Revenue ($)')

# (1,1) Chart type chosen: Pie Chart | Why: Best for showing proportion/share of parts to a whole (few categories)
axes[1, 1].pie(cat_revenue, labels=cat_names, autopct='%1.1f%%', startangle = 90)
axes[1, 1].set_title('Revenue Share by Product Category')

plt.tight_layout()
plt.show()
```

```
# 🤔 Wrap-Up Reflection

# 1. Yes, the core analytical question defines the relationship (trend, ranking, distribution), which naturally dictates the visual encoding before examining raw values.
# 2. Human eyes evaluate linear bar lengths along a shared axis much faster and more accurately than comparing similar angular slice areas in a pie chart.
# 3. Matters: Bar Chart (Task 1 & 7) because non-zero baselines distort visual proportions. Doesn't matter: Scatter Plot (Task 4) or Line Plot (Task 2), where relative t
# 4. The Scatter Plot (Orders vs Revenue) was hardest to choose because it evaluates a dual-variable relationship rather than simple single-variable aggregation.
```

